Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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1.

Let $X_1, \cdots, X_n$ be i.i.d. Bernoulli variables with success probability θ, when $n > 2$, and let $T = \sum_{i=1}^{n} X_i$. Derive the conditional distribution $X_1, \cdots, X_n$ given $T = t$. Since $X_1, \cdots, X_n \overset{i.i.d.}{\sim} Bernoulli(\theta)$, and $T = \sum_{i=1}^{n} X_i \sim Binomial(n, \theta)$ $\mathrm{P}(X_1 = x_1, \cdots, X_n = x_n) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i}$
$P(X_1 = x_1, \cdots, X_n = x_n, T = \sum_{i=1}^{n} X_i = t) = \theta^t(1-\theta)^{n-t}$
$P(X_1 = x_1, \cdots, X_n = x_n | \sum_{i=1}^{n} X_i = t) = \frac{\theta^t(1-\theta)^{n-t}}{\binom{n}{t}\theta^t(1-\theta)^{n-t}}$

$= \frac{1}{\binom{n}{t}}$

2.

Suppose X_1 and X_2 are iid $Poisson(\theta)$ random variables and let $T = X_1 + 2X_2$.

[label=()]Find the conditional distribution of (X_1, X_2) given $T = 7$. For $\theta = 1$ and $\theta = 2$, respectively, calculate all probabilities in the above conditional distribution and present the two conditional distributions numerically.

[label=()]Since $X_1, X_2 \overset{i.i.d.}{\sim} Poisson(\theta)$, then we have $\{X_1 + 2X_2 = 7\} = \{(X_1 = 1, X_2 = 3), (X_1 = 3, X_2 = 2), (X_1$ and $(X_1 = 1, X_2 = 3), (X_1 = 3, X_2 = 2), (X_1 = 5, X_2 = 1), (X_1 = 7, X_2 = 0)$ are mutually exclusive, then $\mathrm{P}(T = 7) = P(X_1 = 1, X_2 = 3) + P(X_1 = 3, X_2 = 2)$
$+ P(X_1 = 5, X_2 = 1) + P(X_1 = 7, X_2 = 0)$
$= \frac{\theta}{1}e^{-\theta} \cdot \frac{\theta^3}{3!}e^{-\theta} + \frac{\theta^3}{3!}e^{-\theta} \cdot \frac{\theta^2}{2!}e^{-\theta} + \frac{\theta^5}{5!}e^{-\theta} \cdot \frac{\theta^1}{1!}e^{-\theta} + \frac{\theta^7}{7!}e^{-\theta} \cdot e^{-\theta}$
$= \frac{\theta^4 e^{-2\theta}}{6}\left(1 + \frac{\theta}{2} + \frac{\theta^2}{20} + \frac{\theta^3}{840}\right)$ Then the conditional distribution of (X_1, X_2) given $T = 7$ is
$\mathrm{P}(X_1 = 1, X_2 = 3 | T = 7) = \frac{P(X_1=1, X_2=3)}{P(T=7)}$
$= \frac{\frac{\theta^4 e^{-2\theta}}{6}}{\frac{\theta^4 e^{-2\theta}}{6}\left(1 + \frac{\theta}{2} + \frac{\theta^2}{20} + \frac{\theta^3}{840}\right)}$
$= \frac{840}{840 + 420\theta + 42\theta^2 + \theta^3}$
$P(X_1 = 3, X_2 = 2 | T = 7) = \frac{P(X_1=3, X_2=2)}{P(T=7)}$
$= \frac{\frac{\theta^4 e^{-2\theta}}{6} \cdot \frac{\theta}{2}}{\frac{\theta^4 e^{-2\theta}}{6}\left(1 + \frac{\theta}{2} + \frac{\theta^2}{20} + \frac{\theta^3}{840}\right)}$
$= \frac{420\theta}{840 + 420\theta + 42\theta^2 + \theta^3}$
$P(X_1 = 5, X_2 = 1 | T = 7) = \frac{P(X_1=5, X_2=1)}{P(T=7)}$
$= \frac{\frac{\theta^4 e^{-2\theta}}{6} \cdot \frac{\theta^2}{20}}{\frac{\theta^4 e^{-2\theta}}{6}\left(1 + \frac{\theta}{2} + \frac{\theta^2}{20} + \frac{\theta^3}{840}\right)}$
$= \frac{42\theta^2}{840 + 420\theta + 42\theta^2 + \theta^3}$
$P(X_1 = 7, X_2 = 0 | T = 7) = \frac{P(X_1=7, X_2=0)}{P(T=7)}$
$= \frac{\frac{\theta^4 e^{-2\theta}}{6} \cdot \frac{\theta^3}{840}}{\frac{\theta^4 e^{-2\theta}}{6}\left(1 + \frac{\theta}{2} + \frac{\theta^2}{20} + \frac{\theta^3}{840}\right)}$
$= \frac{\theta^3}{840 + 420\theta + 42\theta^2 + \theta^3}$ The conditional distribution of $(X_1, X_2)|T = 7$ is given in table 1.

Tabel 1: Conditional distribution of (X_1, X_2) given $T = 7$

$P(X_1 = x_1, X_2 = x_2 \mid T = 7)$	$(x_1 = 1, x_2 = 3)$	$(x_1 = 3, x_2 = 2)$	$(x_1 = 5, x_2 = 1)$	$(x_1 = 7, x_2 = 0)$
$\theta = 1$	$\frac{840}{1303}$	$\frac{420}{1303}$	$\frac{42}{1303}$	$\frac{1}{1303}$
$\theta = 2$	$\frac{840}{1856}$	$\frac{840}{1856}$	$\frac{169}{1856}$	$\frac{8}{1856}$

2.

3.

Let $X_1, \cdots, X_n$ be i.i.d. random variables with mean μ and variance σ^2. Let $\bar{X}$ denote the sample mean and $V = \sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2$.

[label=()]Derive the expected value of $\bar{X}$ and V. Further suppose that $X_1, \cdots, X_n$ are normally distributed. Let $A_{n\times n} = ((a_{ij}))$ be an orthogonal matrix whose first row is $(\frac{1}{\sqrt{n}}, \cdots, \frac{1}{\sqrt{n}})$ and let $Y = AX$, where $Y = (Y_1, \cdots, Y_n)'$ and $X = (X_1, \cdots, X_n)$ are (column) vectors. (It is not necessary to know a_{ij} for $i = 2, \cdots, n$, $j = 1, \cdots, n$ for answering the following questions.)[label=()]

2. (a) Find $\sum_{j=1}^{n} a_{ij}$ for $i = 1, \cdots, n$ and show that $\sum_{i=1}^{n} Y_i^2 = \sum_{i=1}^{n} X_i^2$ (Use properties of orthogonal matrices.)

(b) Express $\bar{X}$ and V in terms (or as functions) of $Y_1, \cdots, Y_n$.

(c) Use (only) *transformation of variables* approach to find the joint distribution of $Y_1, \cdots, Y_n$. Are $Y_1, \cdots, Y_n$ independently distributed and what are their marginal distributions?

(d) Prove that $\bar{X}$ and V are independent given their marginal distributions.

[label=()]Since $E[X_i] = \mu$ and $Var[X_i] = \sigma^2$ for $i = 1, \cdots, n$ $\mathrm{E}[\bar{X}] = E\left[\frac{\sum_{i=1}^{m} X_i}{n}\right] = \frac{\sum_{i=1}^{n} E[X_i]}{n} = \frac{n\mu}{n} = \mu$ $\mathrm{Var}\left[\bar{X}\right] = E\left[\left(\bar{X} - \mu\right)^2\right] = Var\left[\frac{\sum_{i=1}^{n} X_i}{n}\right]$
$= \frac{\sum_{i=1}^{n} Var[X_i]}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$ $\mathrm{E}[V] = E\left[\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2\right] = E\left[\sum_{i=1}^{n}\left((X_i - \mu) - (\bar{X} - \mu)\right)^2\right]$
$= E\left[\sum_{i=1}^{n}\left(X_i - \mu\right)^2\right] + nE\left[\left(\bar{X} - \mu\right)^2\right] - 2E\left[\sum_{i=1}^{n}(X_i - \mu)(\bar{X} - \mu)\right]$
$= E\left[\sum_{i=1}^{n}\left(X_i - \mu\right)^2\right] - nE\left[\left(\bar{X} - \mu\right)^2\right]$
$= nVar[X_i] - nVar[\bar{X}]$
$= n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2$ Or since $\mathrm{E}\left[\bar{X}^2\right] = Var[\bar{X}] + E[\bar{X}^2]$
$= \frac{\sigma^2}{n} + \mu^2$ $\mathrm{E}[V] = E\left[\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2\right] = E\left[\sum_{i=1}^{n} X_i^2 - 2\sum_{i=1} X_i\bar{X}^2 + n\bar{X}^2\right]$
$= E\left[\sum_{i=1}^{n} X_i^2 - n\bar{X}^2\right]$
$= n \cdot \left[\sigma^2 + \mu^2\right] - n \cdot \left[\frac{\sigma^2}{n} + \mu^2\right]$
$= (n-1)\sigma^2$ [label=()]

2. (a) Due to the orthogonality of A, $A'A = AA' = I_{n\times n}$ ($I_{n\times n}$ is diagonal matrix of 1's. Let $A = (a_{1\cdot}, \cdots, a_{n\cdot})'$ where $a_{j\cdot}$ is the jth row vector. Then we have for $i, j = 1, \cdots, n$

$$a_{i\cdot}a_{i\cdot}' = 1 \text{ and } a_{i\cdot}a_{j\cdot}' = 0$$

$\mathrm{a}_{1\cdot}a_{1\cdot}' = \sum_{k=1}^{n} \frac{1}{\sqrt{n}} \cdot \frac{1}{\sqrt{n}} = \frac{\sum_{j=1}^{n} a_{1j}}{\sqrt{n}} = 1$

$a_{i\cdot}a_{1\cdot}' = \sum_{j=1}^{n} a_{ij} \cdot \frac{1}{\sqrt{n}} = \frac{\sum_{j=1}^{n} a_{ij}}{\sqrt{n}} = 0$

Hence $\sum_{j=1}^{n} a_{ij} = \sqrt{n}$ for $j = 1$ and $\sum_{j=1}^{n} a_{ij} = 0$ for $j = 2, \cdots, n$.

$\sum_{i=1}^{n} Y_i^2 = Y'Y = X'A'AX = X'(A'A)X$

$= X'X = \sum_{i=1}^{n} X_i^2$

(b) Note that $Y_1 = \sum_{i=1}^{n} \frac{1}{\sqrt{n}} \cdot X_i = \frac{\sum_{i=1}^{n} X_i}{\sqrt{n}} = \sqrt{n} \cdot \bar{X}$

$\sum_{i=2}^{n} Y_i^2 = Y'Y - Y_1^2 = \sum_{i=1}^{n} X_i^2 - \left(\sqrt{n}\bar{X}\right)^2$

$= \sum_{i=1}^{n} X_i - n\bar{X}^2$

$= \sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2$ Therefore $\bar{X} = \frac{Y_1}{\sqrt{n}}$ and $\sum_{i=1}^{n} \left(X_i - \bar{X}\right)^2 = \sum_{i=2}^{n} Y_i^2$

(c) since $X_1, \cdots, X_n \overset{i.i.d.}{\sim} N\left(\mu, \sigma^2\right)$

$\mathrm{f}_{X_1,\cdots,X_n}(x_1, \cdots, x_n) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x_i-\mu)^2}{2\sigma^2}}$

$= (2\pi)^{-\frac{n}{2}} \sigma^{-n} e^{-\frac{\sum_{i=1}^{n}(x_i-\mu)^2}{2\sigma^2}}$

$= (2\pi)^{-\frac{n}{2}} \sigma^{-n} e^{-\frac{(X-\mathbf{1}\mu)'(X-\mathbf{1}\mu)}{2\sigma^2}}$

where $\mathbf{1} = (1, \cdots, 1)'$. Let $A = (a_{\cdot 1}, \cdots, a_{\cdot n})$, $A' = (a_{\cdot 1}, \cdots, a_{\cdot n})'$ where $a_{\cdot j}$ is the jth column vector, since $Y = AX$, $X = A'AX = A'Y$, $\frac{d}{dY}X = A$, $\left|\frac{d}{dY}X\right| = \det(A) = \det(A'A) = 1$, then we have

$\mathrm{f}_{Y_1,\cdots,Y_n}(y_1, \cdots, y_n) = f_{X_1,\cdots,X_n}(x_1, \cdots, x_n) \left|\frac{d}{dY}X\right|\Big|_{X=A'Y}$

$= (2\pi)^{-\frac{n}{2}} \sigma^{-n} e^{-\frac{(X-\mu)'(X-\mu)}{2\sigma^2}}\Big|_{X=A'Y}$

$= (2\pi)^{-\frac{n}{2}} \sigma^{-n} e^{-\frac{(A'Y-\mathbf{1}\mu)'(A'Y-\mathbf{1}\mu)}{2\sigma^2}}$

$= (2\pi)^{-\frac{n}{2}} \sigma^{-n} e^{-\frac{(Y-A\mathbf{1}\mu)'AA'(Y-A\mathbf{1}\mu)}{2\sigma^2}}$

$= (2\pi)^{-\frac{n}{2}} \sigma^{-n} e^{-\frac{(Y-A\mathbf{1}\mu)'(Y-A\mathbf{1}\mu)}{2\sigma^2}}$

$= (2\pi)^{-\frac{n}{2}} \sigma^{-n} e^{-\frac{\sum_{i=1}^{n}(y_i - a_{i\cdot}\mathbf{1}\mu)^2}{2\sigma^2}}$

$= (2\pi)^{-\frac{n}{2}} \sigma^{-n} e^{-\frac{\sum_{i=1}^{n}\left(y_i - \sum_{j=1}^{n} a_{ij}\mu\right)^2}{2\sigma^2}}$

$= \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(y_1-\sqrt{n}\mu)^2}{2\sigma^2}} \cdot \prod_{i=2}^{n} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{y_i^2}{2\sigma^2}}$

The last equation is due to (b) (i). Note that $E[Y] = AE[X] = A\mathbf{1}\mu$,$Var[Y] = A'Var[X]A = A'A\sigma^2 = I \cdot \sigma^2$, therefore $Y_1 \sim N(\sqrt{n}\mu, \sigma^2) \perp Y_2, \cdots, Y_n \overset{i.i.d.}{\sim} N(0, \sigma^2)$

Or

$\sum_{i=1}^{n}(X_i - \mu)^2 = \sum_{i=1}^{n} X_i^2 - 2\mu\sum_{i=1}^{n} X_i + n\mu^2$

$= \sum_{i=1}^{n} Y_i^2 - 2\sqrt{n}Y_i + n\mu^2$

$= \sum_{i=2}^{n} Y_i^2 + (Y_1 - \sqrt{n}\mu)^2$

The second equation is due to (b) (i). Hence

$$\mathrm{f}_{Y_1,\cdots,Y_n}(y_1,\cdots,y_n) = f_{X_1,\cdots,X_n}(x_1,\cdots,x_n)\left|\frac{d}{dY}X\right|\Big|_{X=A'Y}$$
$$= \frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(y_1-\sqrt{n}\mu)^2}{2\sigma^2}} \cdot \prod_{i=2}^{n}\frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{y_i^2}{2\sigma^2}}$$

(d) Since $Y_1 \sim N(\sqrt{n}\mu,\sigma^2) \perp Y_2,\cdots,Y_n \overset{i.i.d.}{\sim} N(0,\sigma^2)$, $\bar{X} = \frac{Y_1}{\sqrt{n}} \sim N\left(\mu,\frac{\sigma^2}{\sqrt{n}}\right) \perp Y_2,\cdots,Y_n$ and $\frac{Y_2^2}{\sigma^2},\cdots,\frac{Y_n^2}{\sigma^2} \overset{i.i.d.}{\sim} \chi_1^2$, then $\sum_{i=2}^{n}\frac{Y_i^2}{\sigma^2} \sim \chi_{n-1}^2$. Therefore $\bar{X} \sim N\left(\mu,\frac{\sigma^2}{\sqrt{n}}\right) \perp V = \sum_{i=2}^{n}Y_i^2 \sim \sigma^2\cdot\chi_{n-1}^2$

4.

Consider a large population of individuals and let θ denote the (unknown) proportion of the population belonging to a sensitive group A (e.g. drug users).

Suppose, we randomly select n individuals from the population and ask each person to select a card from a deck and answer the question written on the card. Each card in the deck has one of the two questions: Q_1: Do you belong to A? and Q_2: Do you not belong to A? Also, 85% percent of the cards ask Q_1 and the remaining 15% ask Q_2.

Assume that each person answers Yes or No truthfully to the selected question. For $i = 1,\cdots,n$, let $X_i = 1$ if the ith person answers 'Yes' otherwise $X_i = 0$. So, the data are the observed values of $X_1,\cdots,X_n$.

Give the joint distribution of $X_1,\cdots,X_n$ and the distribution of the total number of Yes responses.

We first consider to calculate the probability for the ith person to answer 'Yes'

$\mathrm{P}(\mathrm{X}_i = 1) = \quad P(answer\ Q_1)\cdot P\left('Yes'\ as\ response|\ answer\ Q_1\right)$
$+\ P(answer\ Q_2)\cdot P\left('Yes'\ as\ response|\ answer\ Q_2\right)$
$= 0.85\times\theta + 0.15\times(1-\theta)$
$= 0.15 + 0.7\theta$ Then we have $X_1,\cdots,X_n \overset{i.i.d.}{\sim} Bernoulli\,(0.15+0.7\theta)$, therefore

$\mathrm{P}(X_1 = x_1,\cdots,X_n = x_n) = \prod_{i=1}^{n}(0.15+0.7\theta)^{x_i}(1-(0.15+0.7\theta))^{1-x_i}$
$= (0.15+0.7\theta)^{\sum_{i=1}^{n}x_i}(0.85-0.7\theta)^{n-\sum_{i=1}^{n}x_i}$

Let $Y_n = \sum_{i=1}^{n}X_i$ be the total number of 'Yes' response, then $Y_n \sim Binomial\,(n, 0.15+0.7\theta)$

$$\mathrm{P}(Y_n = y) = \begin{pmatrix} n \\ y \end{pmatrix}(0.15+0.7\theta)^y(0.85-0.7\theta)^{n-y} \quad for\ y = 0,\cdots,n$$